

```
In [1]: import pandas as pd
```

```
In [2]: data = pd.read_csv("trends.csv")  
df = pd.DataFrame(data)
```

```
In [3]: df.head()
```

```
Out[3]:
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

```
In [4]: df.tail()
```

```
Out[4]:
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt mồi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
In [5]: df.shape
```

```
Out[5]: (26955, 5)
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 26955 entries, 0 to 26954  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   location    26955 non-null  object  
1   year        26955 non-null  int64  
2   category    26955 non-null  object  
3   rank        26955 non-null  int64  
4   query       26955 non-null  object  
dtypes: int64(2), object(3)  
memory usage: 1.0+ MB
```

```
In [7]: df.describe()
```

Out [7]:

	year	rank
count	26955.000000	26955.000000
mean	2015.243369	3.000000
std	3.564683	1.41424
min	2001.000000	1.000000
25%	2013.000000	2.000000
50%	2016.000000	3.000000
75%	2018.000000	4.000000
max	2020.000000	5.000000

In [8]: `df.columns`

Out[8]: `Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')`

In [9]: `df.dtypes`

Out[9]: `location object
year int64
category object
rank int64
query object
dtype: object`

In [10]: `df.isnull().sum()`

Out[10]: `location 0
year 0
category 0
rank 0
query 0
dtype: int64`

In [11]: `df.duplicated().sum()`

Out[11]: `np.int64(10)`

In [12]: `df.nunique()`

Out[12]: `location 83
year 20
category 2450
rank 5
query 18431
dtype: int64`

In [17]: `df.drop_duplicates(inplace = True)`

In [18]: `df.duplicated().sum()`

Out[18]: `np.int64(0)`

